

Recognise and use combinations of transformations to create tessellations and other geometric patterns, using dynamic geometric software where appropriate

Learning intention

- recognise and use combinations of transformations to create tessellations and other geometric patterns, using dynamic geometric software where appropriate.

Teach and model

- understanding that the horizontal coordinate is written first and is changed if there is a move to the left or right.
- using the Cartesian plane to draw lines and polygons, listing co-ordinates in the correct order to complete a polygon.
- understanding that the Cartesian plane provides a graphical or visual way of describing location with respect to a fixed origin.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.